Introduction to Artificial intelligence

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Outline

- What is AI
- Topics covered in this course
  - Knowledge representation
  - Fuzzy logic
  - Searching algorithms
  - Machine learning
  - Deep learning

Meet HAL

- *2001: A Space Odyssey*
  - classic science fiction movie from 1969
- Part of the story centers around an intelligent computer called HAL
- HAL is the “brain” of an intelligent spaceship
- in the movie, HAL can
  - speaks easily with the crew
  - sees and understands the emotions of the crew
  - navigates the ship automatically
  - diagnoses on-board problems
  - makes life-and-death decisions
  - displays emotions
- In 1969 this was science fiction: is it still science fiction?

Different Types of Artificial Intelligence

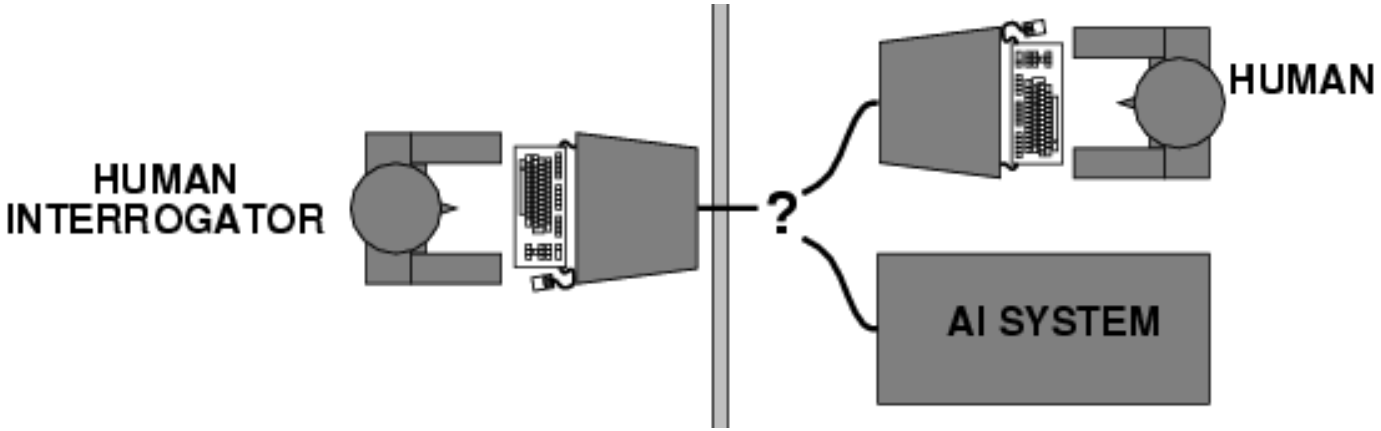
- Modeling exactly how humans think
  - cognitive models of human reasoning
  - Planning, solving puzzles, playing chess, math formulas, optimization (approximations, not just the optimal solutions)
- Modeling exactly how humans act
  - models of human behavior (what they do, not how they think)
  - Shopping, how a student moves around a university campus
    - human actions are often influenced by underlying cognitive processes. However, the primary focus is on the actions themselves.

Cont.

- Modeling how ideal agents “should think”
  - models of “rational” thought (formal logic), reasoning, expert systems
  - note: humans are often not rational! Many things can influence humans' thinking (anger, love)
- Modeling how ideal agents “should act”
  - Rational actions but not necessarily formal rational reasoning
  - i.e., more of a black-box/engineering approach, based on experience, machine learning)
  - E.g., self-driving vehicles
- Modern AI focuses on the last definition
  - success is judged by how well the agent performs
  - modern methods are also inspired by cognitive & neuroscience (how people think).

Acting humanly: Turing Test

- Turing (1950) "Computing machinery and intelligence":
- "Can machines think?" → "Can machines behave intelligently?"
- Operational test for intelligent behavior: the Imitation Game
- Suggested major components of AI:
  - knowledge representation
  - reasoning,
  - language/image understanding,
  - learning



Acting rationally: rational agent

- Rational behavior: Doing what is expected to maximize a “utility function” in a given environment.
- An agent is an entity that perceives and acts.
- A rational agent acts rationally.
- This course is about designing rational agents

- Abstractly, an agent is a function that maps percept histories to actions:

$$[f: \mathcal{P}^\star \rightarrow \mathcal{A}]$$

- For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance
- Note: computational limitations make perfect rationality unachievable

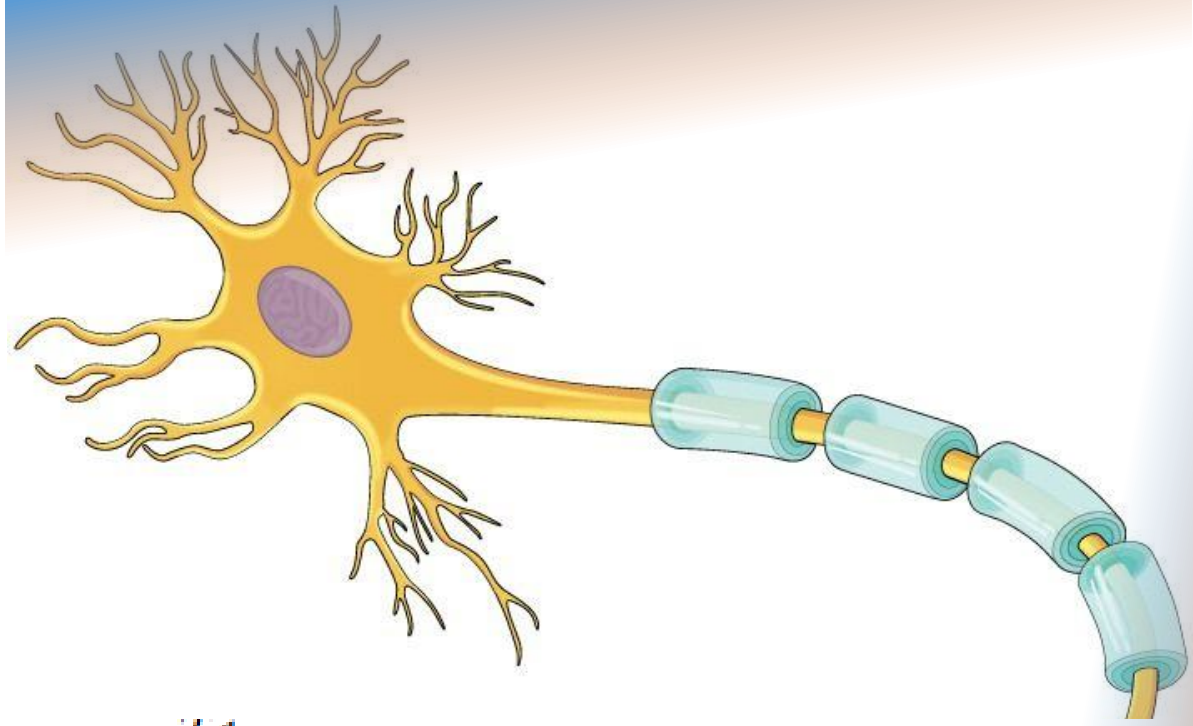
→ design the best program for given machine resources

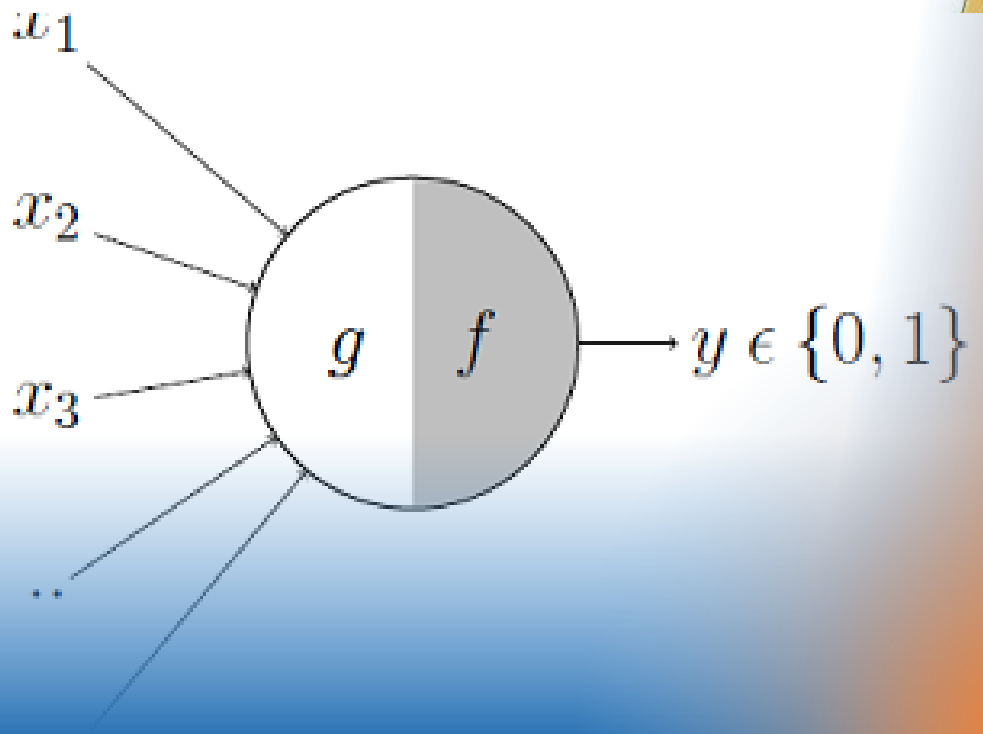
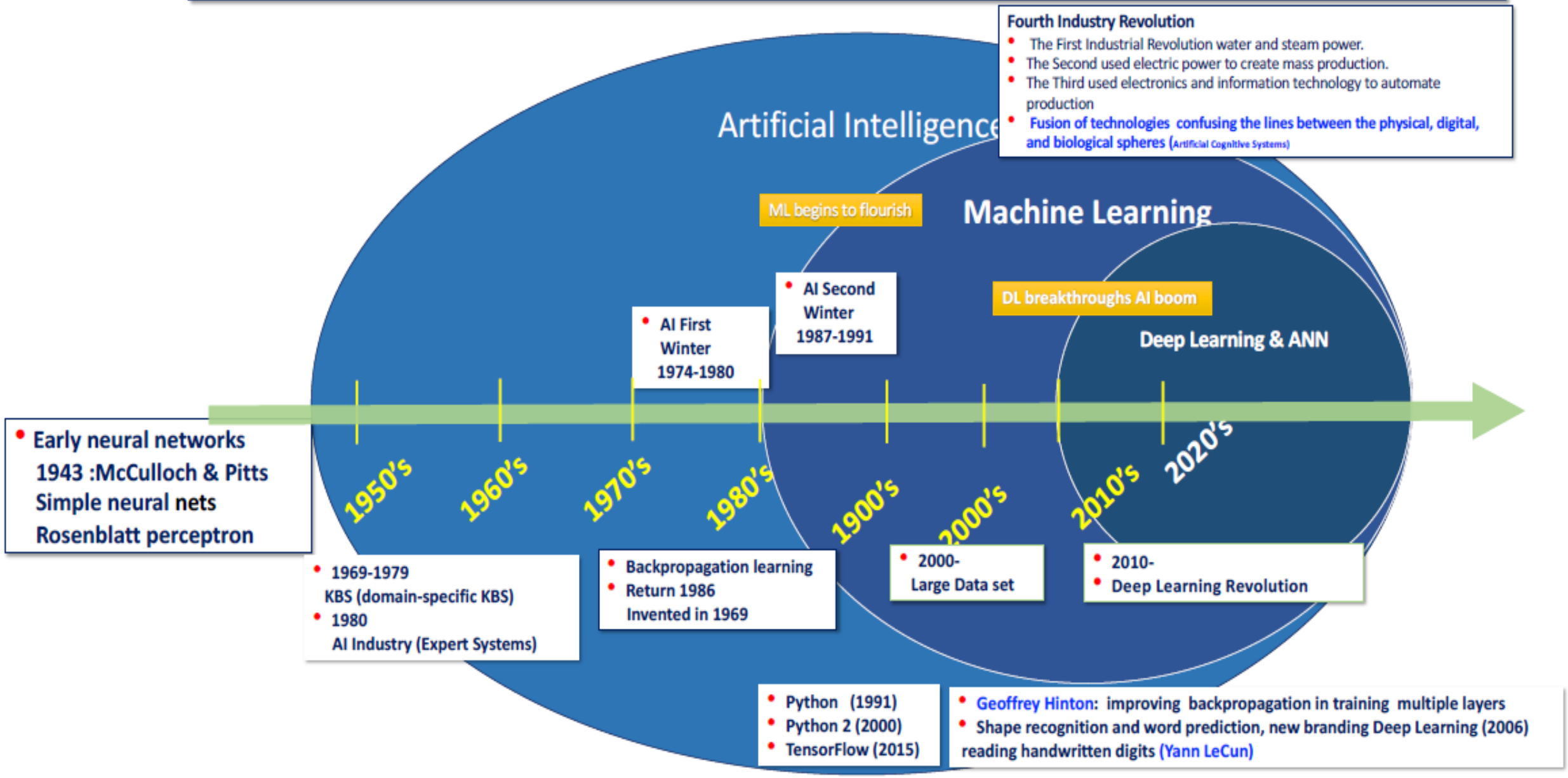
Academic Disciplines important to AI.

- **Philosophy** Logic, methods of reasoning, mind as physical system, foundations of learning, language, rationality.
- **Mathematics** Formal representation and proof, algorithms, computation, probability.
- **Economics** utility, decision theory, rational economic agents
- **Neuroscience** neurons as information processing units.
- **Psychology/** how do people behave, perceive, process Cognitive Science information, represent knowledge.
- **Computer engineering** building fast computers
- **Control theory** design systems that maximize an objective function over time
- **Linguistics** knowledge representation, grammar

History of AI

- 1943 McCulloch & Pitts: Boolean circuit model of brain
- 1950 Turing's "Computing Machinery and Intelligence"
- 1956 Dartmouth meeting: "Artificial Intelligence" adopted
- 1950s Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist,
- 1965 Robinson's complete algorithm for logical reasoning (resolution principle)
- 1966—73 discovering computational complexity (Cook's theorem) Neural network research almost disappears
- 1969—79 Early development of knowledge-based systems
- 1980-- AI becomes an industry
- 1986-- Neural networks return to popularity (Backpropagation popularity)
- 1987-- AI becomes a science
- 1995-- The emergence of intelligent agents
- 2011-- Deep learning revolution (2012 AlexNet)





- Artificial Neural Networks (ANNs) is a subset of AI
- ANN has the capacity to solve problems such as regression and classification with high levels of accuracy.